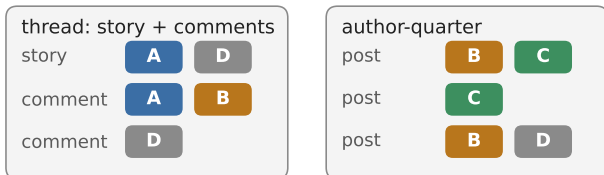
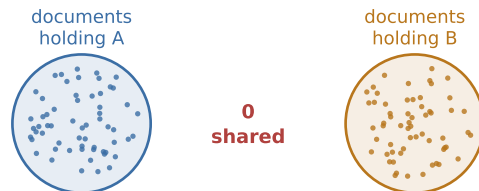


a Documents: a thread or an author-quarter



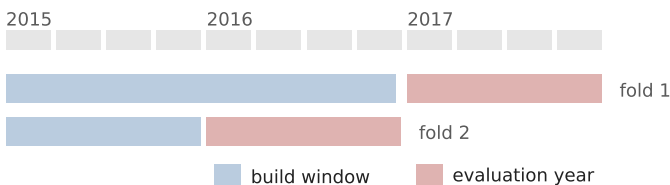
Two document definitions. A thread is a story with its top comments; an author-quarter is every concept one person used in three months. Two concepts co-occur when one document holds both.

b An eligible pair: should have met, never did



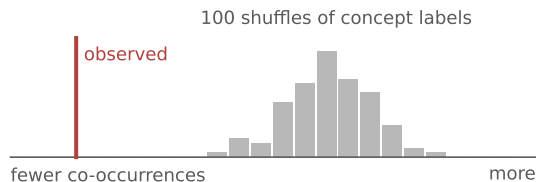
Both concepts are frequent. Under independence, their expected joint count is at least 2 build-window documents. Observed: none. The pair is eligible, or suppressed: it was expected to co-occur but never did.

c Two folds: build on past years, evaluate the next



Eligible pairs are found in the build window; both readouts are measured in the evaluation year. Fold 1 builds on 2015 to 2016 and evaluates 2017; fold 2 builds on 2015 and evaluates 2016. Every fold boundary was registered before the evaluation ran.

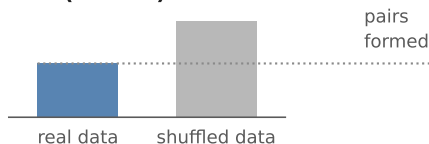
d Segregation: observed counts vs. label shuffles



Sum document counts separately for every eligible pair. Compare with 100 label shuffles, then deduplicate within documents. Binary margins can change. $z = (\text{observed} - \text{null mean}) / \text{null SD}$. Negative z means fewer joint occurrences.

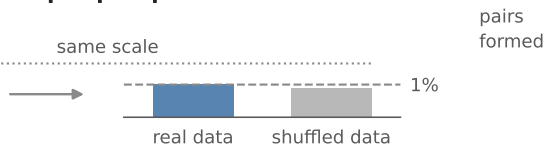
e Two criteria for 'this pair formed', and what shuffled data do under each

z-criterion (retired)



Expected count from each concept's own frequency. Large documents make joint counts combinatorially; shuffled data 'form' more pairs than the real data.

per-pair permutation criterion



Each pair's threshold is its shuffled 99th percentile. Dashed line: nominal 1 percent count reference, not a calibrated error rate or a guaranteed count bound.